

Why forgetting is the *price of remembering cheaply.*

Honest version: Transformers are that friend who saves every screenshot. Great memory, dead phone storage. Recurrent models are the friend who keeps one whiteboard and wipes it. We kept arguing about which friend BDH is. Spoiler: it picked the whiteboard — on purpose. Come break it with us.

OUR BET • PROVE US WRONG

KV-cache: one drawer per word. Grows forever, never lies. **Whiteboard: N slots, total.** Cheap forever, quietly lies when two words want the same corner. If you can keep the whiteboard honest at 500 words in 16 slots, we lose.

Built by Cloud 9, DataForge. If you know attention a little, you're good. 6 mins. The toy below is ours — scrappy, live, definitely not the real BDH.

HOW THIS PAGE WORKS

No lecture. Just mess with it.

1 • STARE

It's already running — 120 pairs, 32 slots. Cabinet's chilling at 100%. Whiteboard's already messed up ~80 times. Click a yellow cell. Ask it what happened.

2 • BE MEAN

Press "Flood it". That's 500 words into 16 slots — hostel-wifi-energy. Then flip the **state rule**: hash vs mamba-gate vs delta vs hola vs bdh-p. Same flood, five different ways to die.

3 • OH, SO BDH DID THIS TOO

Yeah. The real machine chose the whiteboard life. Same headache, fancier math. Scroll and we'll show you the receipts.

THE LAB • LIVE TOY

Two memories, one stream of garbage words.

We feed both the same junk — mango-3 → 42, chai-7 → 88, you get it. Cabinet files everything. Whiteboard goes $\text{hash}(\text{key}) \% N$ and whoever wrote last wins — like a mess register where the last entry counts. Pick a key below and catch the whiteboard lying.

PAIRS STREAMING IN • N



120 pairs

WHITEBOARD SLOTS • N

32 ▼

STATE RULE • 5 WAYS TO PAY

hash — last wins

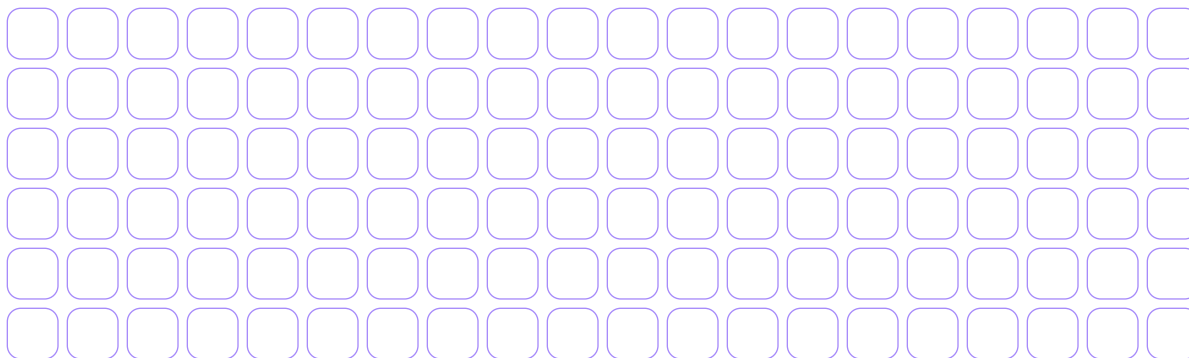
INTERROGATE KEY

▼ lotus-0 ▼

hash: cartoon of raw overwrite. Last writer wins the slot. Honest baseline — real recurrent states interfere by vector superposition ($S += v k^T$), not modulo hashing. Same invariant: $n > \text{capacity} \rightarrow$ silent degradation. See Judge bridge below. Cartoons of write/read rules for teaching — not real kernels. Toy $\neq$ official models.

Filing cabinet — KV-cache $O(n)$ • exact

Slots held: 120



Grows one per token. Retrieval: dot-product over everything. Cost: you pay for every old token, forever.

Whiteboard — recurrent state 0(1) · lossy

Slots: 32 · overwrites: 88



Yellow = someone got erased here. Red = currently lying to you. Click a cell to cross-examine.

Truth beside estimate

Key	Truth	KV says	Whiteboard says	Who stole it?
lotus-0 drawer 9 · hash	65	65	84 ✗	overwritten by orbit-5, rickshaw-3, monsoon-2

Rule hash recall: 26.7% · Cabinet: 100% · Slot collisions: 88

Judge bridge · hash % N → real superposition: Softmax attention stores keys explicitly ($K \in \mathbb{R}^{n \times d}$). Recurrent linear attention / BDH accumulates additively: $S_t = S_{t-1}U + v_t k_t^T$. Querying gives $q^T S_t = (q^T k_t) v_t + \sum_{\tau < t} (q^T U^{t-\tau} k_\tau) v_\tau$ — the sum is cross-talk when keys are not orthogonal ($n > d$). Our hash maps n keys into N slots as a discrete proxy for projecting n tokens into a d -dim state. Delta-rule (blend) is closest to this: averaging models loss of purity where neither value is verbatim recoverable.

THE BILL · LIVE CURVES

Exactness sends an invoice.

Left: drawers you have to buy. Right: answers you get wrong. Both drawn live from your N — and yeah, they print fine now, we learned that the hard way (canvas, never again).

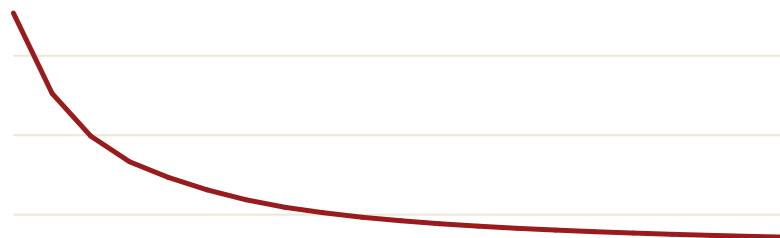
Slots vs tokens

purple climbs $O(n)$ · amber flat $O(1)$

Right now: cabinet **120** drawers vs whiteboard **32** under rule **hash**. At 500: 500 vs 32.

Whiteboard accuracy vs stream length

accuracy per rule – live (hash)



Rule hash: $n=120, N=32 \rightarrow$ **24.1%**. Switch rules at same n/N — that gap is the whole talk.

BDH MODULE · THE PART WE ACTUALLY SWEATED OVER

Okay, so the real machine did the whiteboard thing too.

Who: Dragon Hatchling (BDH), Pathway's weird-and-cool Post-Transformer. Quick note so nobody gets yelled at in Q&A: **not** a Mamba-style SSM — BDH-GPU is its own GPU-friendly rewrite from ReLU low-rank maps + linear attention. BDH doesn't keep drawers; it scribbles into **synapses while reading**. The one equation we're confident enough to frame (Eq.5):

$\rho_{t-1,l} = \sum_{\tau < t} v_{\tau,l-1}^* x_{\tau,l}^T U^{t-\tau}$ – BDH-GPU attention state, Eq.5: fixed-shape matrix, additive updates, $O(1)$ in length

Same deal as your whiteboard upstairs, just with learned keys, a decay U , and ~5% of neurons awake at a time. Why not standard linear attention (Katharopoulos / Performer with fixed ϕ)? Those use fixed positive feature maps to approximate softmax and suffer unnormalized drift. BDH uses strictly non-negative sparse activations via ReLU low-rank maps plus Hebbian plasticity modulated by U – stable $O(1)$ without $O(n^2)$. Full math: Kosowski et al. §3.2 Eqs.4–5, Hebbian lens 6–8, toy code App.E.

Receipt #1 · vendor told us, we didn't verify: 134M BDH gets **95% at 32K → 82% at 128K** on BABILong QA1–QA5 (pathway.com). Holds long, sags under load — exactly your toy's curve, at real scale. Pending contamination / independent / leaderboard review. A benchmark is not a deployment, we'll say it in the viva too.

Receipt #2 · the cousin BDH-CQ: 150M, **29.5% pass@2 on ARC-AGI-1 @ ~\$0.0007/task** (arXiv:2608.09888). Thinks in latent space, no chain-of-thought essays — cheap, not champion. Self-reported, weights unreleased.

Don't leave with the wrong lesson (we almost did): this forgetting is silent — no "sorry, forgot" flag. And remembering for one chat $\neq$ learning forever. Getting fast memory into slow weights is still unsolved. Also please, never call BDH an SSM in front of the judges.

LANDSCAPE IN 30 SECONDS

Who pays what?

Family	Memory	Recall	Bill	Why care
Transformer KV	$O(n)$	perfect	$O(T^2)$, OOMs	gold standard, expensive
Mamba SSM '24	$O(1)$ gated	lossy far	cheap	throughput king, weak needle
HOLA '26	$O(1)$ +cache	recovered	cheap+cache	hippocampus for surprises

Family	Memory	Recall	Bill	Why care
BDH / CQ	O(1) synaptic	lossy, sags	lowest \$/task shown	memory = reasoning fabric

60-SECOND CHECK • BE HONEST

Bet you can't call the next move.

No marks, no tracking. Judges will basically ask you this in the viva, so practice getting it wrong here first.

- ☐ **n=500, N=16:** whiteboard stays perfect because hashing is uniform.
- ☐ **n=500, N=16:** whiteboard collapses; too many keys share too few slots.
- ☐ **KV-cache:** forgets oldest tokens first to stay O(1).

Answer, then set N=16 and flood it to check yourself.

RECEIPTS • VERIFY, DON'T TRUST

Sources & caps.

- Kosowski et al. *Dragon Hatchling: Missing Link between Transformer and Brain*. arXiv:2509.26507 (30 Sep 2025). Eqs.4–8, BDH-GPU App.E. [arxiv](#)
- Engdahl, Kosowski et al. *BDH-CQ: In-Context Learning with Recurrent Latent Reasoning*. arXiv:2608.09888 (10 Aug 2026). $S_t = U(S_{t-1}, D_t)$. [arxiv](#)
- Cui. *Hippocampus for Linear Attention*. arXiv:2607.02303 (2 Jul 2026). O(1) but lossy; overwrites. [arxiv](#)
- Gu & Dao. *Mamba*. 2024. SSM baseline — related, distinct from BDH.

Caps, honestly: toy = string-hash % N, int values, exact-match. No learned keys, decay, sparsity. Small-N curves are live; BABILong/ARC are precomputed vendor numbers. Toy ≠ official BDH. Code: team-written with AI drafting help (Muse Spark) for code, prose, and design; team traced, tested, and defends every line.

Built by Cloud 9 for DataForge — if something looks off, it's probably our hash function, tell us in Q&A.

Single file, no login, no secrets · [repo](#) · [live](#)